ME523 Final Project

Thermodynamics II – Fall 2023

# Contents

# Introduction

The McMurdo base in Antarctica sustains 75 scientists and their research efforts though the harsh environmental climate. We have designed a power cycle to provide electricity, water, and air that will sustain the scientists for their day to day lives.

Using the assumptions provided below:

- The air is “dry” air
- All pipes are “well insulated”
- Isentropic efficiencies cannot eclipse 85%
- Turbines, compressors, and pumps have heat loss that is 4-6% of the work produced or consumed
- Heat exchangers, combustors, and steam generators have stray heat transfer of 8-10%

The system can provide electrical needs of 12 MWhr per person per year. To better understand, the system provides a total of 2.466 MWhr per day for the entire facility. All water is brought up to 65 °C for safety purposes before being split into cooling water and heating water. The system provides 230 liters of water per person per day. In other words, 8,625 liters of both hot and cold water are provided every day for the scientists. 6000 m^3 of air with 0.35 air changes per hour is provided to the facility and constantly maintained between 21 °C and 25 °C for indoor air. All heating air is maintained between 32 °C and 34 °C, while all cooling air is maintained between 7 °C and 10 °C.

In our report, we have provided the finalized diagram of the entire system. This details how the electrical/thermal energy, water, and air system works and how they are connected to each other. Each section of the system is then described in detail with all necessary variables provided. This further explains how we chose and sized each of our components. Schematics of the T-s and P-v diagrams for each section are provided. This provides a further visual of how the working fluids flow between each component. All work and heat transfer calculations are shown in detail in the appendix along with all other calculations. The major safety concerns are reviewed at the end of the report to ensure this system can be safely manufactured and operated.

## Overall Cycle Schematic

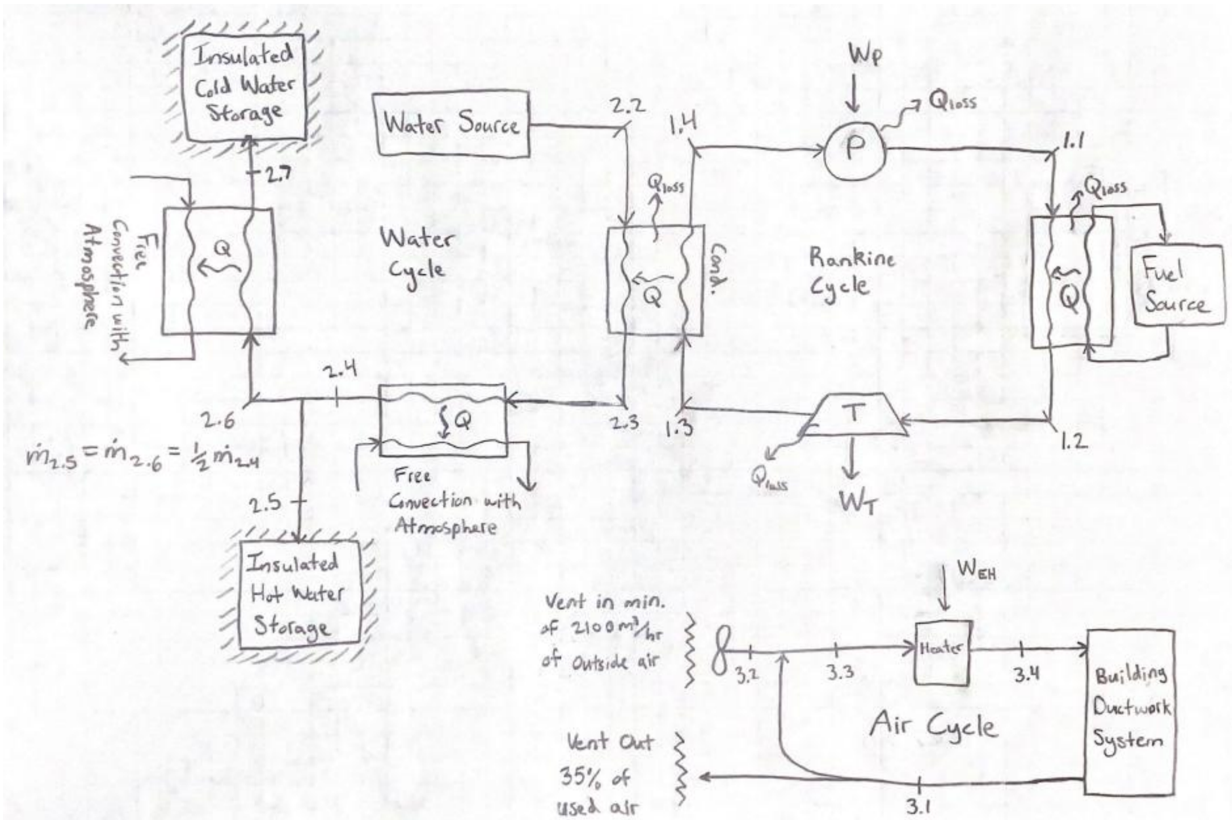


*Figure 1*

## Power Cycle

The power cycle is a simple Rankine cycle that uses water as its working fluid. The output requirements of the power cycle are to provide *at least* 83.3 kW of heating to the water cycle and a net work of 41.5kW to power the air heater and fulfill the scientists' daily electricity needs. The schematic of the system is shown in Figure 3.

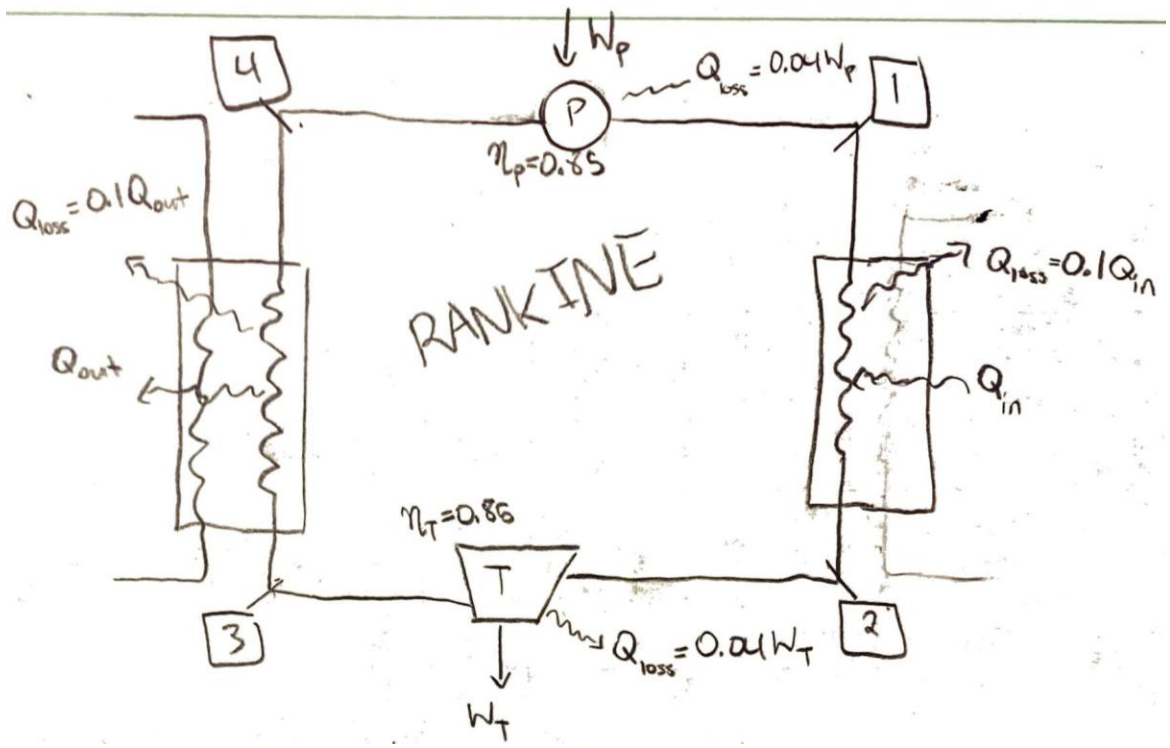


*Figure 2 - Power cycle schematic*

| Power Cycle States | Phase | Pressure (bar) | Temperature (C) | Enthalpy (kJ/kg) |
|---|---|---|---|---|
| 1.1 | Condensed liquid | 200 | 183.45 | 788.386 |
| 1.1s | Condensed liquid | 200 | 182.57 | 761.678 |
| 1.2 | Superheated vapor | 200 | 626.3 | 3537.6 |
| 1.3 | Saturated vapor | 10 | 179.9 | 2778.1 |
| 1.3s | 2-phase | 10 | 179.9 | 2741.17 |
| 1.4 | Saturated liquid | 10 | 179.9 | 762.81 |

*Table 1 - Power cycle states*

The pressures of 200bar and 10bar were set, as well as the state 2 temperature of 600 degrees Celsius. The mass flow rate of the cycle was calculated based on the work of the turbine, which produces the work needed to heat up the air in the HVAC system, power the pump in the Rankine cycle, and provide the necessary power to the 75 scientists (**Error! Reference source not found.**). The mass flow rate calculated to produce the minimum work needed is 0.248 kg/s.

The turbine and pump both have 85% efficiencies. These values, along with the assumption that state 3 is a saturated vapor and state 4 is a saturated liquid, are used to calculate the enthalpy values needed for all work and heat calculations (**Error! Reference source not found.**, **Error! Reference source not found.**).

| Turbine Work ($W_t$) | Pump Work ($W_p$) | Condenser Heat ($Q_{out}$) | Combustion Heat ($Q_{in}$) | Stray Heat Loss |
|---|---|---|---|---|
| **153.968 kW** | **5.176 kW** | **449.81 kW** | **681.805 kW** | **76.672 kW** |

*Table 2 - Power cycle work, heat, and heat loss*

The cycle efficiency is calculated by dividing the net work of the cycle by the heat added (**Error! Reference source not found.**). For this cycle, the net work is 148.522 kW ($W_t$ - $W_p$).

| Cycle Thermal Efficiency (η) |
|---|
| **21.78%** |

### Combustion

For the heat addition into the Rankine cycle, the design will use propane combusted from 255K (-18 °C) to 800K. The chemical equation of the reaction was determined to be $C_3H_8 + 5(O_2+3.76N_2)$ — $3CO_2$ + $4H_20 + 18.8N_2$ (**Error! Reference source not found.**). At these temperatures, the energy from the process is 362868 kJ/$kmol_f$. Given that the Rankine cycle requires 681.805 kW and an additional 10% of energy will be lost to the environment, the necessary amount of propane on an annual basis is **2,873,763 kg** (**Error! Reference source not found.**, **Error! Reference source not found.**).

## Water System

The water system can be seen in the overall diagram in **Error! Reference source not found.**. The water is first pulled into the system and runs through the first of three heat exchangers. The first heat exchanger is the condenser from the Rankine Cycle. This is done to utilize the stray heat transfer of the Rankine Cycle to heat up the inlet water to the minimum required temperature of 65°C. The water exits the first heat exchanger at 65°C and is ready to be cooled back down to the hot water usage temperature of 55°C. This is done by feeding all the water through the second heat exchanger that works using free convection with the environment. The water supply exits the second heat exchanger at 55°C and is then split in half. Half of the water supply goes to a well-insulated “Hot Water Storage Tank” while the other half of the water is sent through the third and final heat exchanger. Once again, this heat exchanger uses free convection with the environment to cool the remainder of the water supply to the cold-water usage temperature of 10°C. The cold water is then sent to a well-insulated “Cold Water Storage Tank”.

The water cycle mass flow rate was calculated based on the information given that each of the 75 scientists needs 230 Liters of water per day (**Error! Reference source not found.**). The mass flow rate of the water cycle through the first two heat exchangers is 0.3318 kg/s. Once the water supply is split in half for the hot water vs cold water storage at states 2.4, 2.5 and 2.6 the mass flow rate is 0.1679 kg/s. Once the mass flow rate was determined, it was then time to calculate the heat transfer for each heat exchanger. The main Q value that we were looking for was the one in the first heat exchanger/Rankine Cycle condenser. It was found that 83.279 kJ/s of energy leaves the Rankine Cycle and enters the Water Cycle (**Error! Reference source not found.**).

The mass flow rate information was also used to calculate the sizing of pipes for each segment of the water cycle. Based on the assumption that the water source comes from a lake that is 200 Meters up in the mountains, that means that the water has a velocity of 63.35 m/s. That was found by setting PE = KE and solving for velocity. It was determined that piping would be needed that ranged in size from 2.61 mm diameter down to 1.83 mm diameter (**Error! Reference source not found.**).

## Air System

For the air system, it is important to highlight the necessary HVAC requirements in both the most extreme conditions and the average temperatures throughout the year. The extreme conditions will be considered for the "design day" to determine the necessary sizing of the heating and cooling components in our system, while the average conditions will indicate the anticipated annual heating load required by the McMurdo's occupants. From knowledge provided by the researchers, the design temperatures are listed in **Error! Reference source not found.** below.

| Low Temperature (°C) | Average Temperature (°C) | High Temperature (°C) |
|---|---|---|
| -50 | -18 | 8 |

*Table 3 - Air cycle design temperature considerations*

<u>Schematic</u>

The design of the HVAC system is labeled in **Error! Reference source not found.** below to handle the air needs of our occupants. States 3.1 and 3.2 account for the indoor and outdoor air respectively. At state 3.3, the air streams will mix at a controlled rate. Once mixed, the air will move through a heater and reach state 3.4 to be released into the building through the ductwork.

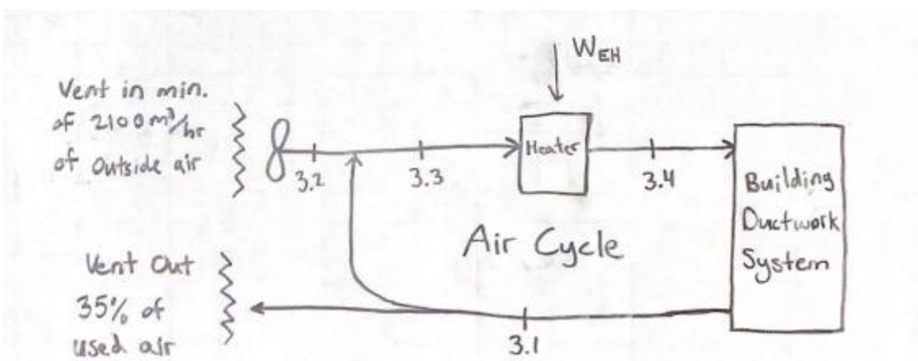


*Figure 3 – HVAC system schematic*

<u>Ventilation</u>

The ventilation requirements at the McMurdo facility are 35% air change on an hour basis, or 2100 $m^3$. This poses challenges in the low energy environment as the temperature difference between the outdoor air and indoor air is so drastic, which in turn requires much more energy to maintain a comfortable internal environment at the base.

<u>Heating</u>

The system will provide heat to the base when the indoor temperature drops to 21 °C or below. To warm the environment our system will have air exiting the vents at 32 °C. To minimize energy use, the system is designed to only warm the required air needed for ventilation purposes. Through calculation A.3 (**Error! Reference source not found.**), the required heating load was determined to be 67.75 kW, or 20 tons at the extreme temperature of -50 °C.

<u>Cooling</u>

Cooling the base will only occur in situations where the indoor temperature reaches or exceeds 25 °C. In these situations, air will exit the vents at 10 °C to bring the internal temperature down. Given the high temperature of 8 °C, the cooling process can be completed through the absence of any energy addition considering that a mix of the 8 °C air and 25 °C air can achieve the desired 10 °C air at the exit of the vents. Calculation A.1 (**Error! Reference source not found.**) demonstrates that proportion of 8 °C and 25 °C air to be 0.8823 and 0.1177 respectively. Therefore, the mixed air will simply remain at 10 °C and atmospheric pressure.

<u>Average Annual Energy Consumption</u>

To determine the total power needs of McMurdo's facility, the annual power consumption of the HVAC system on the base is found based on the average temperature provided of –18 °C. This is considering that the HVAC system will spend nearly the entire year in heating. On an average daily basis, the anticipated average heating load was determined to be 1.0905 MWhr/day (**Error! Reference source not found.**).

<u>Recommendation</u>

For the HVAC equipment to best suit the needs of this facility, our design recommendation is to move forward with a Trane Foundation (20 Ton Electric) Rooftop Unit, which has been included in the visual below. This unit provides the necessary heating load requirement of our HVAC system through electric heat and has the necessary controls to maintain operation at any temperature between our design days. Additionally, the equipment achieves industry leading efficiency and reliability to minimize the costs and potential complications in the years to come.

*Figure 4 - Trane Foundation Rooftop Unit*

# Safety

## Water Cycle

The main safety concern for the water cycle is the possible buildup of microorganisms in the cooling water. Bacteria, fungi, and algae are possible contributors that can lead to what is called a slime deposit. This is caused by soil, aquatic, or airborne microbes entering the system. Enough slime deposit buildup will cause an insulation barrier on the heat exchanger resulting in an instant deterioration of the heat exchanger's efficiency. To safely provide water to the 75 scientists, all the water must reach a minimum temperature of 65 C, according to the project description. We were able to fulfil this need by installing multiple heat exchangers. All of the water is heated to the required minimum temperature in the first heat exchanger before being split into cold water and hot water. Bringing the water up to this temperature allows us to reduce initial slime deposit buildup. Because slime deposits increase metal corrosion, all cooling water is demineralized and chemically treated. The biggest preventative regarding microbial growth is consistent and thorough checks of the cooling water reservoir.

## Air System

Our air cycle utilizes propane gas as the working fluid, requiring a few safety regulations. Propane is highly flammable, and any sort of leak can lead to an explosion or fire. Proper storage and distribution of propane becomes vitally important. There are two standards that are required when using propane gas via the National Fire Protection Association (NFPA 54 and NFPA 58). These require the system to provide proper pressure of the propane tank and proper leak testing.

## Power Cycle

We wanted to eliminate as many harmful agents as possible, so we utilized a Rankine Cycle with water as the working fluid. Using water allows us to eliminate any possible emissions, such as carbon monoxide, that come with using other refrigerants that can cause danger to the system and its environment. With that, the main concern becomes keeping the water from freezing in the pipes if the system were to shut down. Although the pipes are "well insulated," most well insulated pipes will still freeze below –6 C. This becomes an issue in the case of a power outage as the average temperature is –18 C in McMurdo. We have prevented this by providing an extra layer of foaming insulation across all pipes.

# Conclusion

Our ultimate goals were to provide:

- Electrical needs of 12 MWhr per person per year for the 75 scientists.
- 230 liters of water per person per day (half hot, half cold)
- 6000 m^3 of air with 0.35 air changes per hour

We did this by producing the proper Rankine cycle for power production along with supplying a Train Foundation rooftop which provides the proper amount of power and efficiency for the air system. The water cycle issues were solved by using multiple heat exchangers to combine all the water and divide it into two separate sections with proper temperatures.

We believe that the system we have provided for the 75 scientists will allow them to function fully and live comfortably whilst conducting their research.

# References

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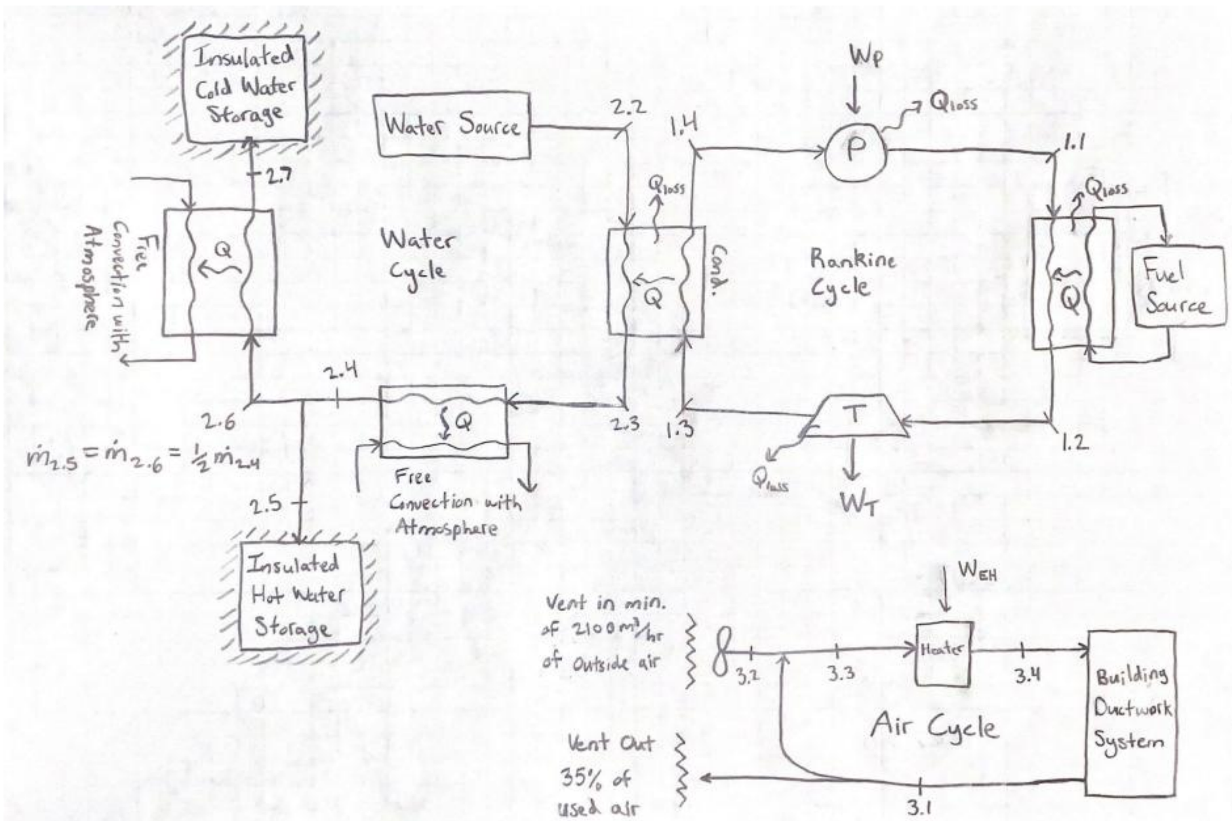
# Appendix

## Figures



Power needs: 75 people 12 MWhr per person per year

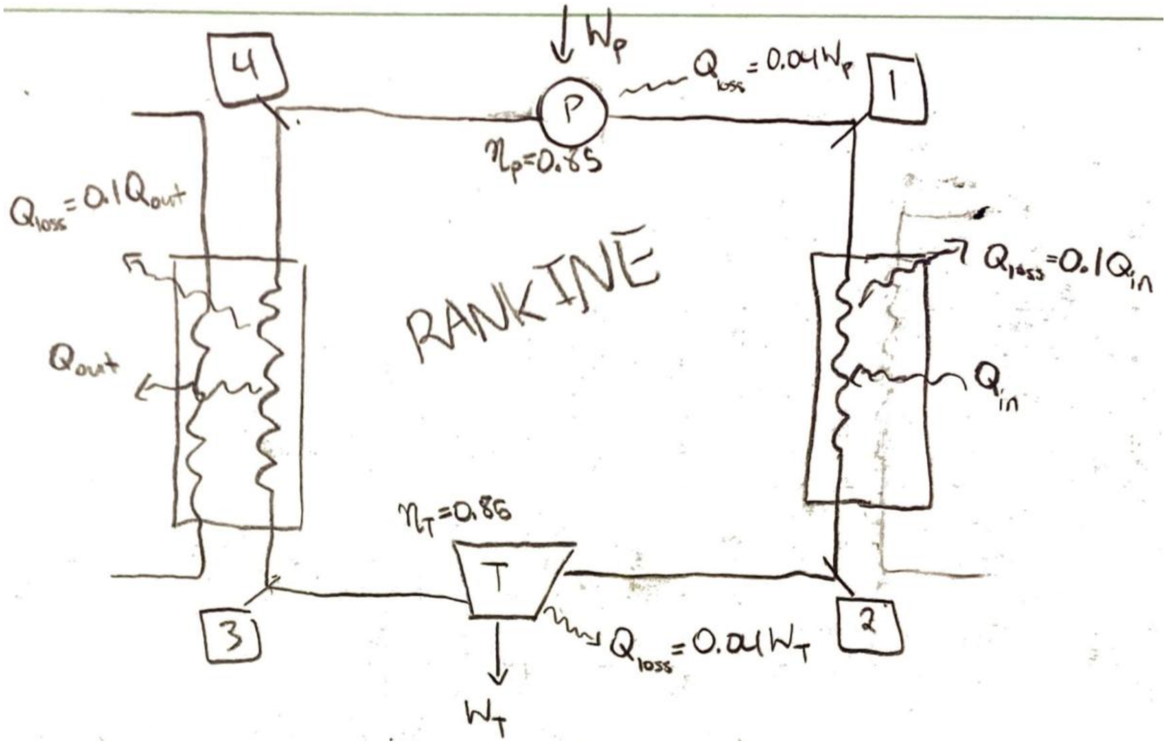
Per year: $75(12) = 900$ MWhr/year

Per Day: $900 \text{ MWhr/year} \left| \frac{1 \text{ year}}{365 \text{ days}} \right| = 2.466$ MWhr/day

Air Needs: $6000\ m^3$ 0.35 air change per hour

New Air: $6000(0.35) = 2100\ m^3/hr$

Reused Air: $6000(0.65) = 3900\ m^3/hr$



$$W_T = W_P + W_{EH} + 2.466 \text{ MWhr/day}$$

$$W_P = \dot{m}(h_4 - h_1)\eta_P \qquad W_{EH} = 1.090584 \text{ MWhr/day}$$

$$W_T = \dot{m}(h_2 - h_3)\eta_T$$

$$\dot{m}(h_2 - h_3)\eta_T = \dot{m}(h_4 - h_1)\eta_P + (1.090584 + 2.466) \text{ MWhr/day}$$

$$\dot{m} = \frac{3.556884 \text{ MWhr/day}}{0.96(h_2 - h_3)\eta_T - 0.96(h_4 - h_1)\eta_P} \qquad \dot{m} = 0.248 \text{ kg/s}$$

account for heat loss

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$S_4 = S_{1S}$, interpolate to find $h_{1S} = 784.55$ kJ/kg

$$\eta_P = \frac{h_{1S} - h_4}{h_1 - h_4} \Rightarrow h_1 = 788.386 \text{ kJ/kg} \quad T_1 = 183.45°C$$

$$W_P = \dot{m}(788.386 - 762.81)\eta_P \cdot 0.96 = 5.176 \text{ kW}$$

$h_2 = 3537.6$ kJ/kg, $h_3 = 2778.1$ kJ/kg

$$W_T = \dot{m}(h_2 - h_3)\eta_T \cdot 0.96 = 153.698 \text{ kW}$$

$$W_{net} = W_T - W_P - \left(3.556884 \frac{\text{MWhr}}{\text{day}}\right) \cdot \frac{1 \text{ day}}{86400 \text{ seconds}} \cdot \frac{3600{,}000 \text{ kJ}}{1 \text{ MWhr}}$$

$$= 153.698 \tfrac{\text{kJ}}{\text{s}} - 5.176 \tfrac{\text{kJ}}{\text{s}} - 148.2035 \tfrac{\text{kJ}}{\text{s}} = 0.3185 \tfrac{\text{kJ}}{\text{s}} > 0$$

The turbine provides enough power

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$$Q_{in} = \dot{m}(h_2 - h_1) = 0.248 \tfrac{\text{kJ}}{\text{s}}(3537.6 - 788.386)\tfrac{\text{kJ}}{\text{kg}} = 681.805 \text{ kJ/s}$$

$$Q_{out} = \dot{m}(h_3 - h_4) \cdot 0.9 = 0.248 \tfrac{\text{kg}}{\text{s}}(2778.1 - 762.81)\tfrac{\text{kJ}}{\text{kg}} = 449.81 \text{ kJ/s}$$

$$Q_{loss_{4\to1}} = \dot{m}(h_1 - h_4)\eta_P - W_P = 0.869 \text{ kW}$$

$$Q_{loss_{2\to3}} = \dot{m}(h_2 - h_3)\eta_T - W_T = 25.823 \text{ kW}$$

$$Q_{loss_{3\to4}} = \dot{m}(h_3 - h_4) - Q_{out} = 49.98 \text{ kW}$$

$$\eta_{cycle} = \frac{W_T - W_P}{Q_{in}} = \frac{153.698 - 5.176}{681.805}$$

Total Heat Loss = 76.672 kW

$$\eta_{cycle} = 21.78\%$$

# Combustion

$$\frac{\dot{Q}}{\dot{m}} = \sum_P n_e(\bar{h}_f^\circ + \Delta\bar{h}) - \sum_R n_i(\bar{h}_f^\circ + \Delta\bar{h})$$

$$C_3H_8 + a(O_2 + 3.76N_2) \rightarrow bCO_2 + dH_2O + eN_2$$

$C: 3 = b$

$H: 8 = 2d \Rightarrow d = 4$

$O: 2a = 2b + d \Rightarrow a = 5$

$N: 3.76a = e \Rightarrow e = 18.8$

Assume:

$T_{ref} = -18°C\ (255K)$

$T_p = 800K$

$$\Rightarrow C_3H_8 + 5(O_2 + 3.76N_2) \rightarrow 3CO_2 + 4H_2O + 18.8N_2$$

| reactants | $n\left(\frac{kmol}{kmol_f}\right)$ | $\bar{h}_f^\circ + \Delta\bar{h}\left(\frac{kJ}{kmol}\right)$ | $n(\bar{h}_f^\circ + \Delta\bar{h})\left(\frac{kJ}{kmol_f}\right)$ |
|---|---|---|---|
| $C_3H_8$ | 1 | $-103850 + 0$ | $-103850$ |
| $O_2$ | 5 | $0 + 0$ | 0 |
| $N_2$ | 18.8 | $0 + 0$ | 0 |
| | | $\Sigma$ | $-103850$ |

products

| | | | |
|---|---|---|---|
| $CO_2$ | 3 | $-169300 + (32179 - 7800)$ | $-434778$ |
| $H_2O$ | 4 | $-104040 + (27896 - 8460.5)$ | $-338418$ |
| $N_2$ | 18.8 | $0 + (23714 - 7412)$ | $306477.6$ |
| | | $\Sigma$ | $-466718$ |

$$\Rightarrow \frac{\dot{Q}}{\dot{m}} = -466718 - (-103850) = -362868\ \frac{kJ}{kmol_f}$$

$$\dot{n}_f\left(\frac{\dot{Q}}{\dot{m}}\right) = Q_{in_{Rankine}}(1 + 0.1)$$

## Combustion

$$\dot{Q}_{in_{Rankine}} = 681.805 \frac{kJ}{s}$$

$$\Rightarrow \dot{n}_f = 681.805 \frac{kJ}{s} (1.1) \left( \frac{1}{362868 \frac{kJ}{kmol}} \right)$$

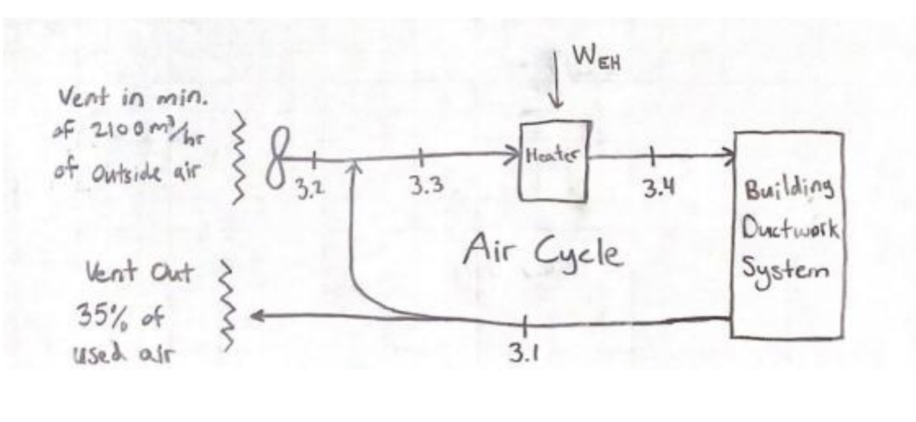
$$\dot{n}_f = 0.0020668 \frac{kmol}{s}$$

$$\dot{m}_f = \dot{n}_f M_f = 0.0020668 \frac{kmols}{s} \left( 44.09 \frac{kg}{kmol} \right)$$

$$\dot{m}_f = 0.091126 \ kg/s$$

On an annual basis:

$$m = 2873763 \, kg$$
of propane



# Air Cycle

**Assume:**

$T_{avg} = -18°C$

$T_{high} = 8°C$

$T_{low} = -50°C$

$6000 m^3$ space

$c_{p_{air}}(255\,K) = 1.0031 \frac{kJ}{kg \cdot K}$

$\rho_{air}(255K) = 1.412 \frac{kg}{m^3}$

35% hourly airchanges

$\dot{V}_{OA} = 0.35(6000m^3) = 2100\,m^3/hr$

**Cooling:** *vents 10°C air*

| OA | RA | Mixed |
|---|---|---|
| 8°C | 25°C | 10°C |

$$\Rightarrow 8x + 25y = 10 \Rightarrow x = 0.8823,\ y = 0.1177 \qquad (A.1)$$

**heating:** *vents 25°C air*

$$\dot{m}_{air} = \dot{V}_{OA} \cdot \rho = 2100\,m^3/hr \left(1.412 \frac{kg}{m^3}\right)\left(\frac{1hr}{3600s}\right)$$

$$\dot{m}_{air} = 0.824\ kg/s \qquad (A.2)$$

@ $T_{low}$)

$$Q = \dot{m}_{air}\, c_{p_{air}} (T_{vents} - (-T_{low}))$$

$$Q = 0.824\ kg/s \left(1.0031 \frac{kJ}{kg \cdot K}\right)(32 - (-50))K$$

$$\Rightarrow Q = 67.75\,kW \qquad (A.3)$$

in tons $\Rightarrow Q = 19.36$ tons $\qquad (A.3)$

@ $T_{avg}$)

$$Q_{avg} = \dot{m}_{air}\, c_{p_{air}} (T_{vents} - (-T_{avg}))$$

$$Q_{avg} = 0.824\ kg/s \left(1.0031 \frac{kJ}{kg \cdot K}\right)(32 - (-18))K$$

$$\Rightarrow Q_{avg} = 41.31\ kW \tag{A.4}$$

Avg daily heating load w/ loss:

$$= Q_{avg}\left(24\ \frac{hr}{day}\right)(1+0.10)$$

$$= 1.0905\ \frac{MWhr}{day} \tag{A.5}$$

## Tables

| Constants | Value | Units |
|---|---|---|
| New air flowrate | 2100 | m^3/hr |
| Electrical energy | 2.466 | MWhr/day |
| Isentropic Efficiency | 0.85 | |
| Average Temp | -18 | C |
| Cold Air Temp | -50 | C |
| Warm Air Temp | 8 | C |
| Cp_air (@ 255 K) | 1.0031 | kJ/kg*K |
| rho_air | 1.412 | kg/m^3 |
| Utilization Rate | 60 | % |

| Power Cycle States | Phase | Pressure (bar) | Temperature (C) | Enthalpy (kJ/kg) |
|---|---|---|---|---|
| 1.1 | Condensed liquid | 200 | 183.45 | 788.386 |
| 1.1s | Condensed liquid | 200 | 182.57 | 761.678 |
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| Turbine Work ($W_t$) | Pump Work ($W_p$) | Condenser Heat ($Q_{out}$) | Combustion Heat ($Q_{in}$) | Stray Heat Loss |
|---|---|---|---|---|
| **153.698 kW** | **5.176 kW** | **449.81 kW** | **681.805 kW** | **76.672 kW** |

| Power Cycle States | (kJ/kg) | kJ/s | MWhr/day |
|---|---|---|---|
| W_41 | 20.870016 | 5.175763968 | 0.124228273 |
| Q_41 (Heat loss) | 0.869584 | 0.215656832 | 0.005176178 |
| Q_12 | 2749.214 | 681.805072 | 16.36463079 |
| W_23 | 619.752 | 153.698496 | 3.689059005 |
| Q_23 (Heat loss) | 25.823 | 6.404104 | 0.153710792 |
| Q_34 | 1813.761 | 449.812728 | 10.79636911 |

| Low Temperature (°C) | Average Temperature (°C) | High Temperature (°C) |
|---|---|---|
| -50 | -18 | 8 |

| Water Needs | | |
|---|---|---|
| 230 Liters per person per day | | |
| 75 people | | |
| 230*75= | 17250 | liters/day |
| 17250*0.001= | 17.25 | m^3/day |
| 17.25/51840= | 0.000332755 | m^3/s |
| 0.0001997*997= | 0.331756366 | kg/s |
| m_dot= | 0.331756366 | kg/s |
| m_dot(2.6_2.7)= | 0.165878183 | kg/s |

| Water Loop States | Q_dot/m_dot (kJ/kg) | Q_dot (kJ/s) |
|---|---|---|
| 2.2-2.3 | 251.08 | 83.29738831 |
| 2.3-2.4 | 41.83 | 13.87736878 |
| 2.6-2.7 | 188.22 | 31.22159158 |

| Water Loop | m_dot(kg/s) | Temp(C) | v(m^3/kg) | A(m^2) | D(m) | D(mm) |
|---|---|---|---|---|---|---|
| 2.2 | 0.33176 | 5 | 1.00E-03 | 5.24E-06 | 0.0026 | 2.58 |
| 2.3 | 0.33176 | 65 | 1.02E-03 | 5.34E-06 | 0.0026 | 2.61 |
| 2.4 | 0.33176 | 55 | 1.01E-03 | 5.31E-06 | 0.0026 | 2.60 |
| 2.5 | 0.16588 | 55 | 1.01E-03 | 2.66E-06 | 0.0018 | 1.84 |
| 2.6 | 0.16588 | 55 | 1.01E-03 | 2.66E-06 | 0.0018 | 1.84 |
| 2.7 | 0.16588 | 10 | 1.00E-03 | 2.62E-06 | 0.0018 | 1.83 |
| Velocity | 63.35 | m/s | | | | |